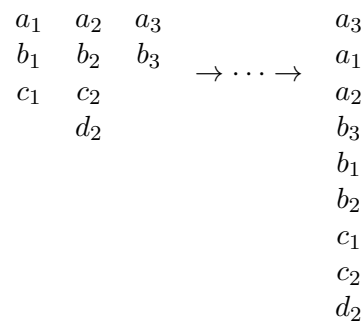


Test #5 of Erlang

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1. This is about shuffling three stacks in the following way (the top of the stacks is on the top of the figure):



If the three stacks are empty, then the result is empty too. Write a function `three/3` computing this kind of shuffle.

2. Write a function `pre/2` (*prefix*) such that, if

$$\text{pre}(P, Q) \stackrel{+}{\rightarrow} R,$$

and list P is the beginning of list Q then R is `true`, otherwise `false`. For example,

$$\begin{aligned}\text{pre}([0, 1, 1, 0, 1], [0, 1, 1, 0, 1, 1, 1, 0, 0]) &\stackrel{+}{\rightarrow} \text{true}, \\ \text{pre}([\text{t}, \text{h}, \text{e}], [\text{t}, \text{h}, \text{e}, \text{s}, \text{e}]) &\stackrel{+}{\rightarrow} \text{true}, \\ \text{pre}([4, \text{a}, 3], [4, \text{a}]) &\stackrel{+}{\rightarrow} \text{false}, \\ \text{pre}([], [1, 2, 3]) &\stackrel{+}{\rightarrow} \text{true}.\end{aligned}$$

Note that the empty list is the prefix of *any* list.